

LLM-as-a-judge for Explanations of NLP models

Master's Thesis, P1, P2, DAP

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1. Background and Motivation

Different explanation mechanisms have been proposed, to show *why* an AI model produced a certain output. Take for example a movie review, and the sentiment prediction “Negative”:

“This was not too bad for a German movie.”

One type of explanations could highlight important *features* that led to the prediction (e.g. “bad”, “German”). Another type of explanation could point to training examples that were influential on the model for making that prediction (“Really bad even for a low-budget movie” -> “Negative”).

There have been different approaches to evaluate the quality of explanations (Nauta et al. 2023). One of the most interesting ways is to test whether humans learned something about the AI model when seeing explanations (Hase & Bansal, 2020), but human experiments are expensive and time consuming. Recently, the idea has emerged to use LLMs instead of humans for evaluating explanations (Wang et al. 2025), but their evaluation only measures *plausibility* (how convincing an explanation appears) and *accuracy* (how much an explanation helps to predict the label), and the study is only done for feature-based explanations and a non-NLP task. The aim of this project is to explore LLM-as-a-judge for different kinds of explanations and NLP tasks.

2. Goals

Depending on the project type (BSc / MSc thesis / DAP / P1 / P2) **one or two** of the following work packages will be worked on:

WP1: Evaluate **feature explanation** methods with a **one phase** LLM-as-a-judge task. Here, the LLM judge sees the input (e.g. a movie review) and an explanation in terms of features (e.g. words such as “great”), and needs to predict the model output. (This is compared to the case where the LLM judge predicts the output from only the input.) Feature explanation methods can be LIME (Ribeiro et al. 2016) or SHAP (Lundberg and Lee 2017). Task datasets can be sentiment prediction or natural language inference tasks (Sun et al. 2025).

WP2: Evaluate **feature explanation** methods with a **two phase** LLM-as-a-judge task. Similar to (Hase & Bansal 2020), the LLM judge sees several examples of inputs, outputs, and explanations, and needs to predict the model output for an input without an explanation. Datasets same as above (Sun et al. 2025).

WP3: Evaluate **training data explanation** methods with a **one phase** LLM-as-a-judge task. The LLM judge sees the input (e.g. a movie review) and an explanation in terms of influential training examples, and needs to predict the model output. Training examples can be selected by similarity, or influence estimation methods (Hammoudeh & Lowd 2024). Datasets same as above (Sun et al. 2025).

WP4: Evaluate **training data explanation** methods with a **two phase** LLM-as-a-judge task. The LLM judge sees several examples of inputs, outputs, and influential training examples, and needs to predict the model output for an input without an explanation. Datasets same as above (Sun et al. 2025).

Each of the work packages have their own technical challenges, for example how to generate the explanations, and how to present them to the LLM judge. They should be accompanied with detailed quantitative and qualitative analysis.

4. Prerequisites

- Experience in Python
- Basic knowledge of machine learning toolkits (e.g., scikit-learn, huggingface)
- Understanding core concepts from natural language processing and machine learning
- Please read the first 20 pages, i.e., Chapters 1, 2 and 3 of (Lyu et al. 2024) before contacting us.

5. Schedule

(Example, if taken as a Data Analysis Project)

Week	Task Description
1	Defining the project scope, schedule, and initial literature with the advisor
1-2	Literature review, setting up the computational environment

3	Conceptualization of own implementation / of data preparation
4-9	Implementation of methods / data preparation
10-12	Detailed evaluation, quantitative and qualitative error analysis (statistics and examples)
13	Preparing presentation, submission of code
14	Presentation of results
16	Submission of documentation

6. References

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